

AI and Sustainability Global Summit

Friday, July 18th, 2025

Thrive in the AI World

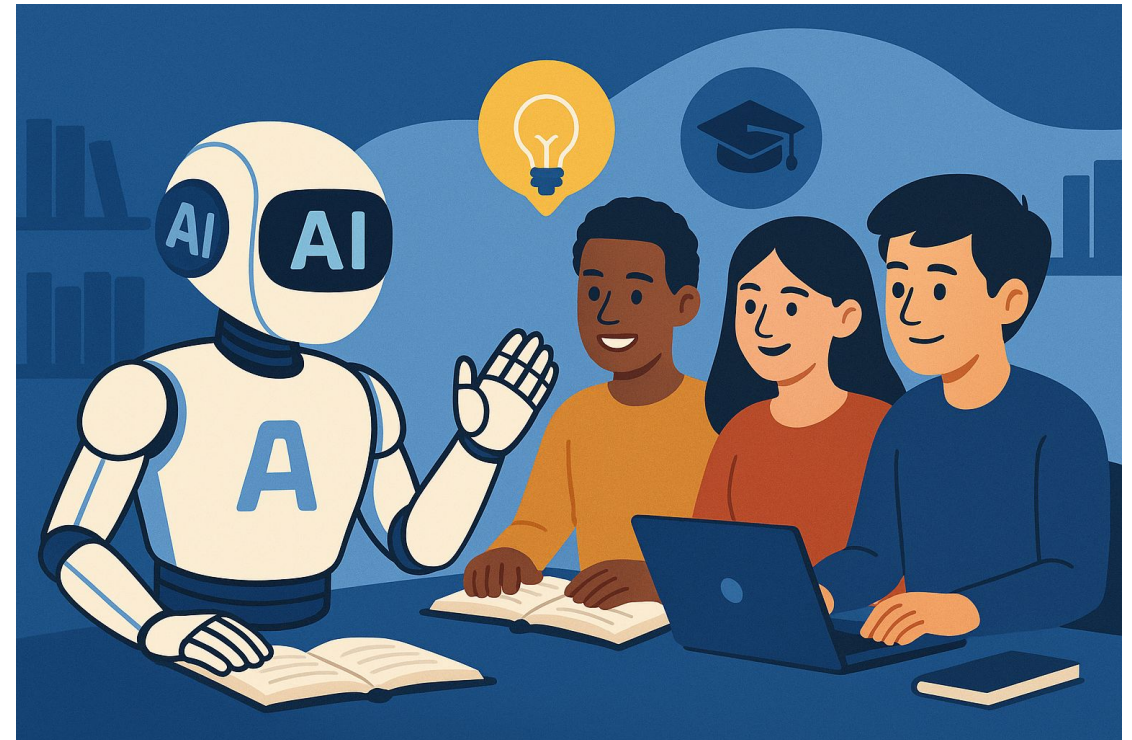
Purpose, guardrails and real-world impact

Session Keynote: AI and Education

Educating the Next Generation of Changemakers
and Impact Leaders: AI is Transforming Education

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Source: Image created with ChatGPT

[Technology] Education Is An Integral Part of the UN SDGs



THE GLOBAL GOALS

TARGET 4.3



EQUAL ACCESS TO AFFORDABLE TECHNICAL, VOCATIONAL AND HIGHER EDUCATION

By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university.

TARGET 4.4



INCREASE THE NUMBER OF PEOPLE WITH RELEVANT SKILLS FOR FINANCIAL SUCCESS

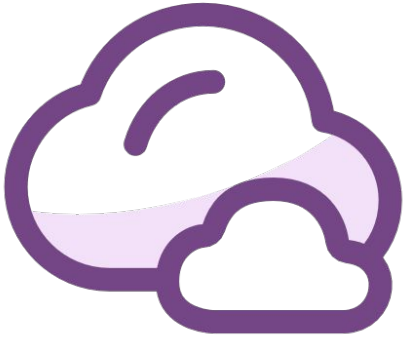
By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship.

4

QUALITY EDUCATION

Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

However, integrating AI in Education has challenges



What do you think is the #1 challenge for educators in higher ed?



ChatGPT

You

Gemini

Rise of Violations against Academic Integrity Through the Gen AI Use. Real-World Examples.

Wrong Type of Apple

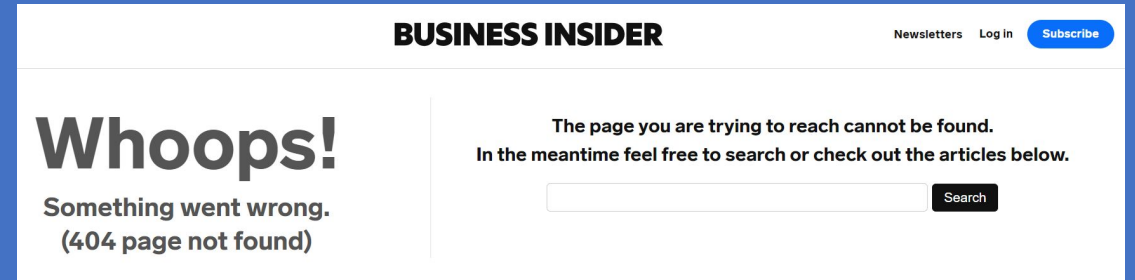
An analysis of the marketing practices of the consumer tech brand Apple uses an article about organic apple harvesting in Chile if references.

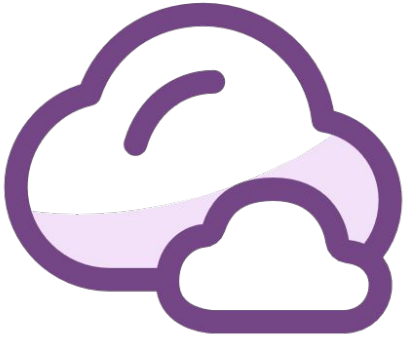
Not Reading the Fine Print

As of my last knowledge update in September 2021, Airbnb had successfully gone public through an initial public offering (IPO) in December 2020, becoming a publicly traded company under the ticker symbol "ABNB" on the NASDAQ (Airbnb, 2021). However, it's important to note that the business landscape can change rapidly, and I recommend checking the most recent sources for the latest developments regarding Airbnb.

Fake Sources

Business Insider. (2022). Nike's Powerful Campaign: Connecting Emotionally with Consumers. Retrieved from <https://www.businessinsider.com/nikes-powerful-campaign-connecting-emotionally-with-consumers>





What is a key opportunity for educators offered by AI?

AI Is Changing What and How We Teach, Learn, and Research



- Increased plagiarism and cheating (Hu, 2023)
- Instructors banned tools, such as ChatGPT (Stringer & Wiggers, 2023)
- Limited protection by plagiarism tools (Dwivedi et al., 2023)
- AI-generated paper submissions to academic journals (Hu, 2023)
- Need to protect authorship of creative works (Livni et al, 2023)



- Personalized learning (Chan & Hu, 2023; Ciampa et al., 2023; Dai et al., 2023).
- Enhanced engagement and motivation through interactive/immersive experiences, adaptive content, gamification, and instant feedback (Naiberh, 2024)
- Enhanced student services with chatbots (Chan & Hu, 2023)
- Accelerate research (OECD, 2023)

Facilitate responsible AI use and development guided by ethics, critical thinking, and academic integrity.

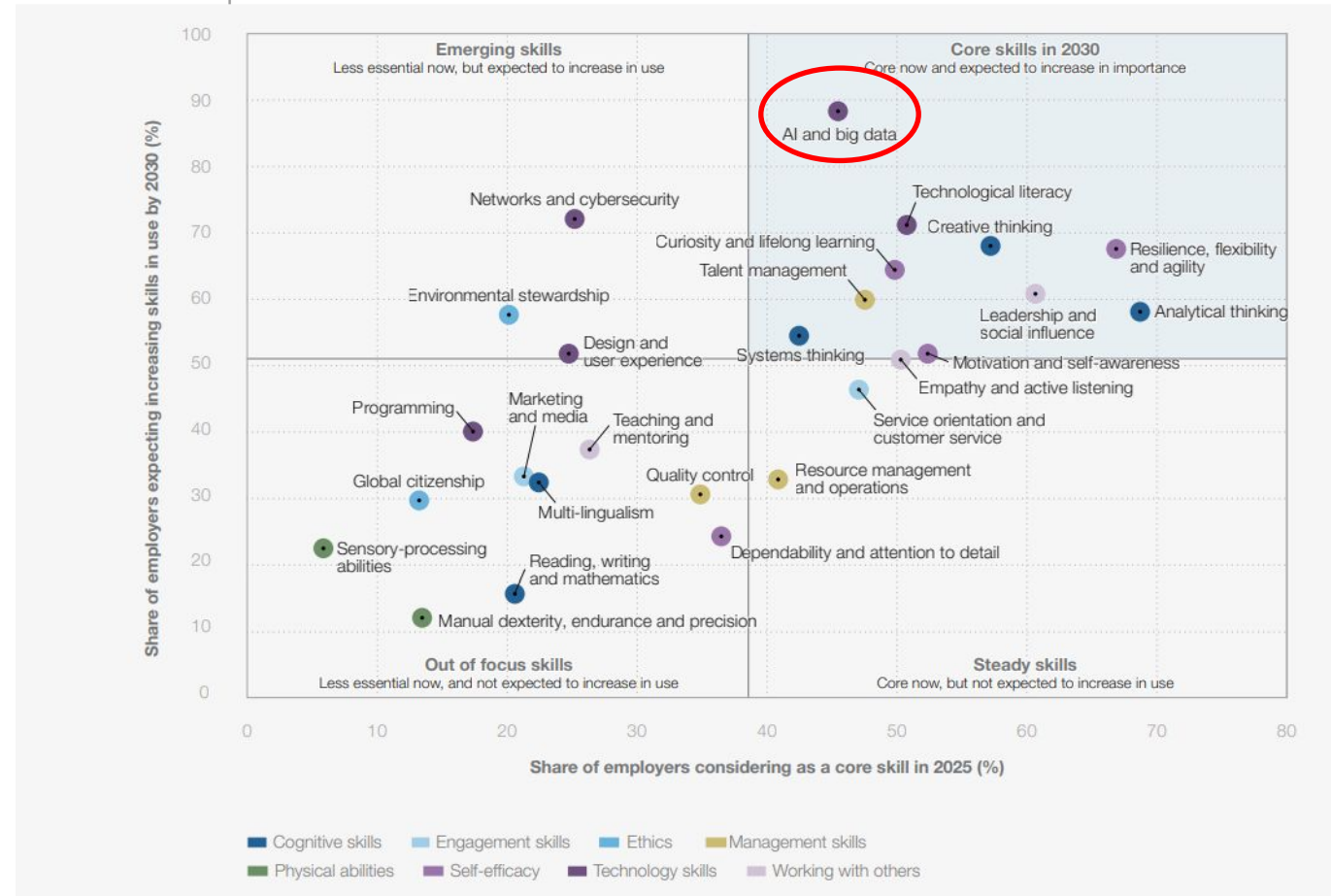
The Evolution of AI Also Changes Which Skills the Industry Expects Graduates to Have

- Digital skills building is critical for careers (Masterson, 2023)
- 88% of U.S. professionals think higher ed should prepare students to use AI before entering the workforce, (SHRM, 2024)
- 74% think incoming college graduates should already have AI user experience prior to entering the workforce.

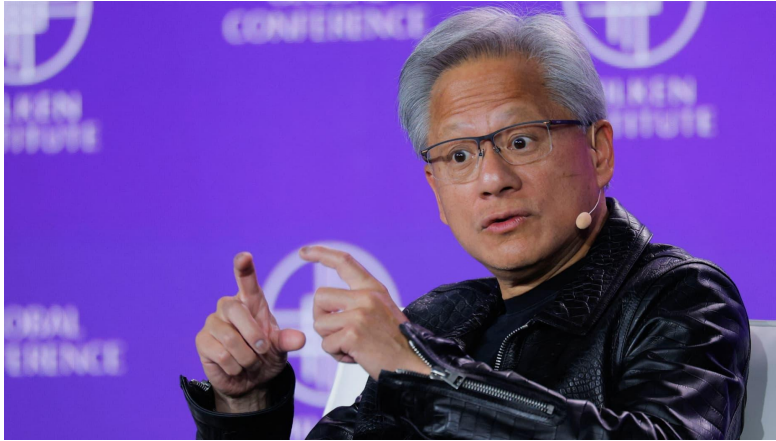
FIGURE 3.6

Core skills in 2030

Share of employers considering skills to be a core skill in 2025 and share of employers expecting skills to increase in importance by 2030.



Industry Views: Need for AI Upskilling



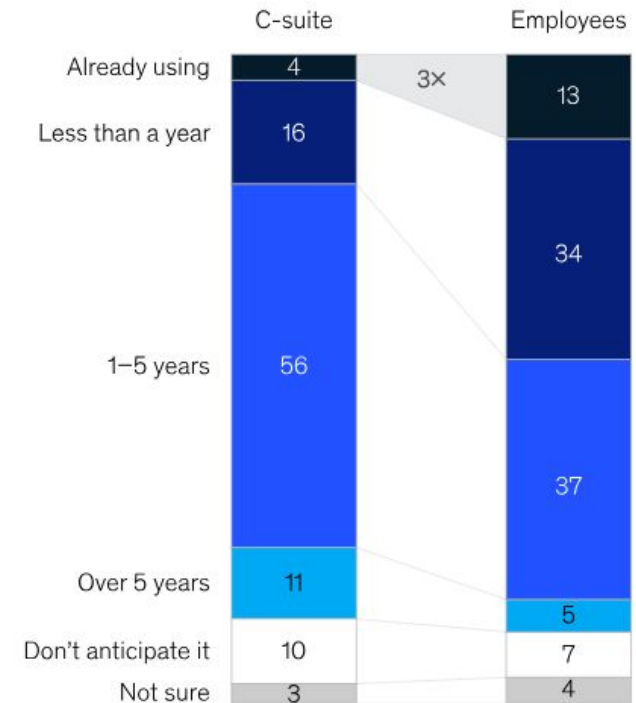
“You’re not going to lose your job to AI — you’re going to lose your job to someone who uses AI.”

- Founder and CEO at Nvidia Jensen Huang at Milken Institute Global Conference 2025 in Beverly Hills, May 5, 2025.

Image credit: Mike Blake, Reuters, published by [CNBC](#)

Employees are three times more likely to be using gen AI today than their leaders expect.

US employees' and C-suite's timeline for employees using gen AI for >30% of daily tasks, % of respondents



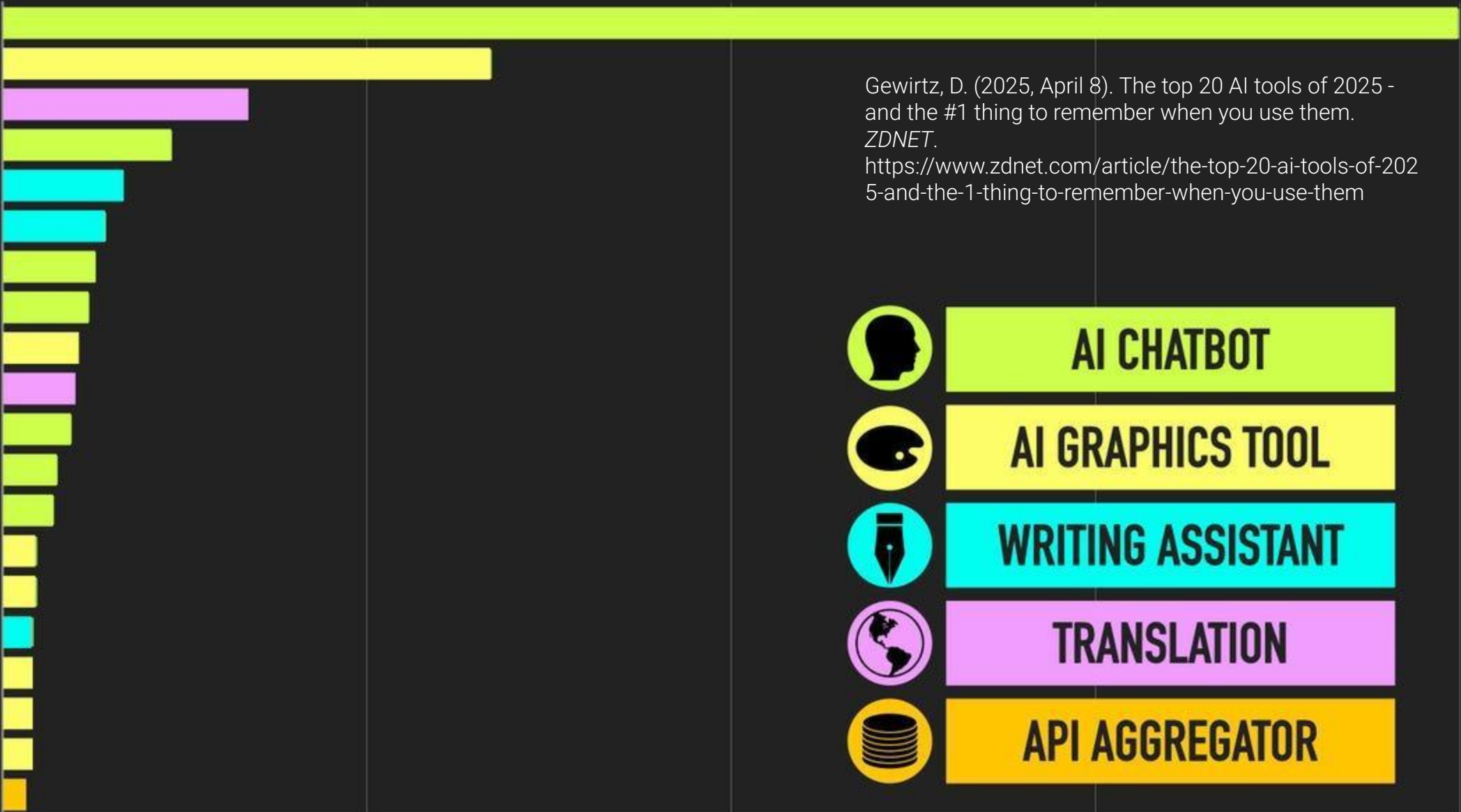
Note: Figures may not sum to 100%, because of rounding.

Source: McKinsey US CxO survey, Oct–Nov 2024 (n = 118); McKinsey US employee survey, Oct–Nov 2024 (n = 3,002)

McKinsey & Company

ZDNET AI TOOL POPULARITY INDEX

CHATGPT
CANVA
DEEPL
GEMINI
GRAMMARLY
QUILLBOT
DEEPSEEK
MS COPILOT
REMOVE.BG
GOOGLE TRANSLATE
PERPLEXITY AI
CHARACTER.AI
CLAUDE
MIDJOURNEY
MS IMAGE CREATOR
JASPER
DALL-E
DREAMSTUDIO
STABLE DIFFUSION
DEEPAI



Gewirtz, D. (2025, April 8). The top 20 AI tools of 2025 - and the #1 thing to remember when you use them. ZDNET.
<https://www.zdnet.com/article/the-top-20-ai-tools-of-2025-and-the-1-thing-to-remember-when-you-use-them>



AI CHATBOT



AI GRAPHICS TOOL



WRITING ASSISTANT



TRANSLATION



API AGGREGATOR

Observations From Teaching



How important is it to learn gen AI tools as part of your studies?

Multiple Choice Poll 12 votes 12 participants

Very Important - 7 votes



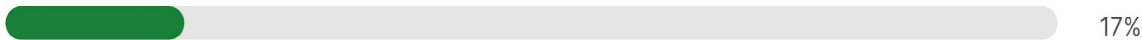
58%

Important - 3 votes



25%

Undecided - 2 votes



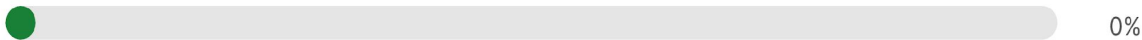
17%

Not that important - 0 votes



0%

Not important at all - 0 votes



0%

Source: Survey in undergraduate marketing class at NYC-based liberal arts college, Spring 2024

- AI adoption was fueled by free accessibility of tools
- Experimenting with tools and insights wanted on how to use each tool
- Now: regionalization of tools: DeepSeek, Le Chat (Mistral AI) and tiering of plans



Which AI tool do you use the most

Student Experiences with AI in Higher Ed

56% students are using AI for tasks such as analyzing data and to produce content or communications (WEF Future of Jobs, 2025)

The State of Permitted Use of AI in Courses at US Academic Institutions (EDUCAUSE, 2025)

Most to all of my instructors are prohibiting student use of generative AI.



52%

Some of my instructors are prohibiting student use of generative AI.



34%

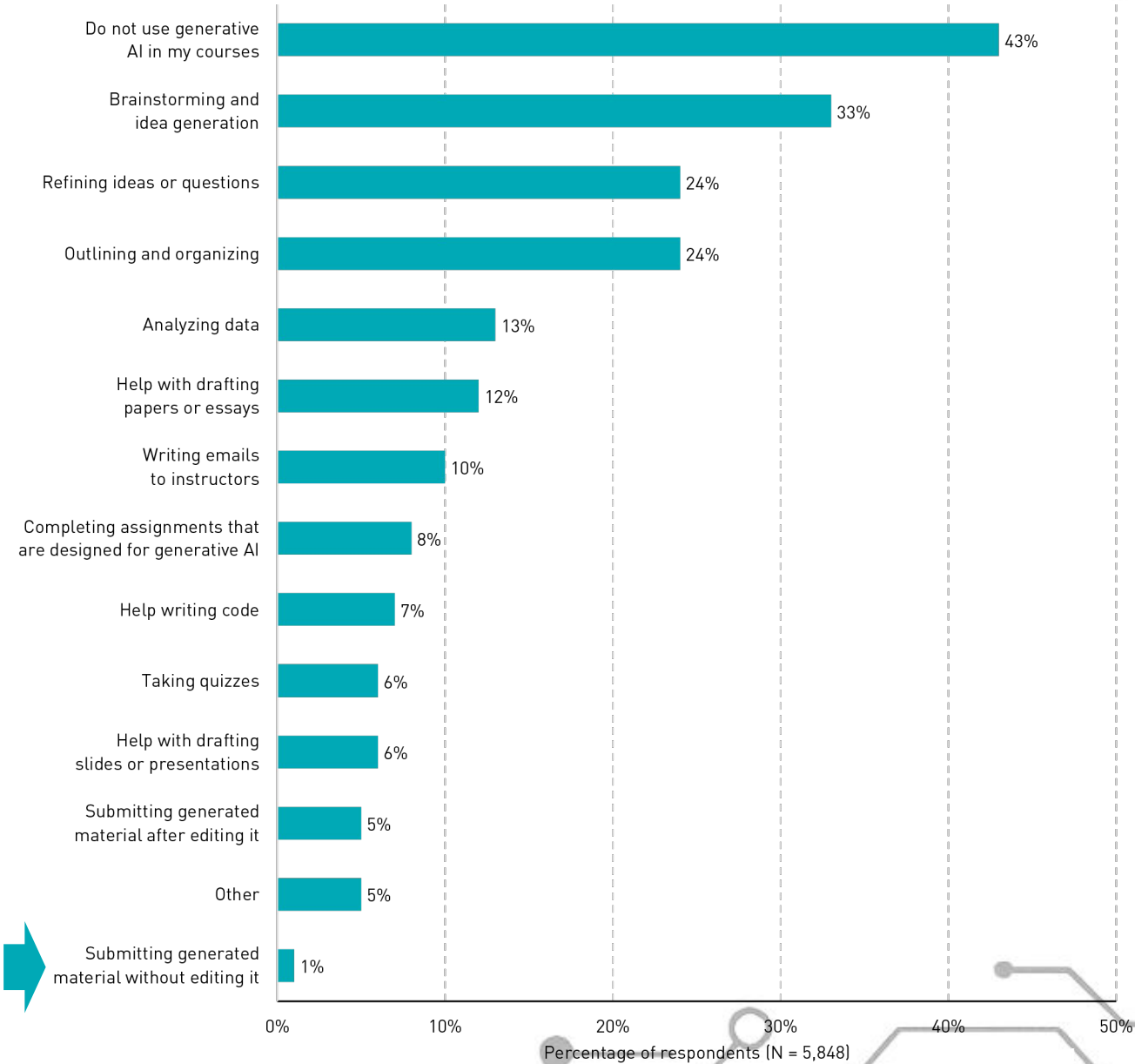
Most to all of my instructors are allowing student use of generative AI.



13%

Percentage of respondents (N = 4,068)

Table 11 in EDUCAUSE 2025 Report: Current Uses of Generative AI in Courses (Muscanell, & Gay, 2025)



Importance of Ground Rules for the Use of AI In Class. Example: [Columbia University AI Policy | Office of the Provost](#)

- Only Use Approved AI Tools/Software. For this class this includes ChatGPT, Claude, Gemini, Meta Imagine, Perplexity,
- Do not input Confidential Information
- Do not input Personal Information
- Do not input information that violates IP or general contract terms and conditions
- Confirm the accuracy of the output provided by Generative AI tools. Check the output of Generative AI tools for bias
- Disclose the use of Generative AI tools
- Comply with third-party intellectual property rights
- Do not use Generative AI tools to produce malicious content
- Only submit your OWN WORK and ensure ACADEMIC INTEGRITY

Level of Preparedness for an AI Work Environment

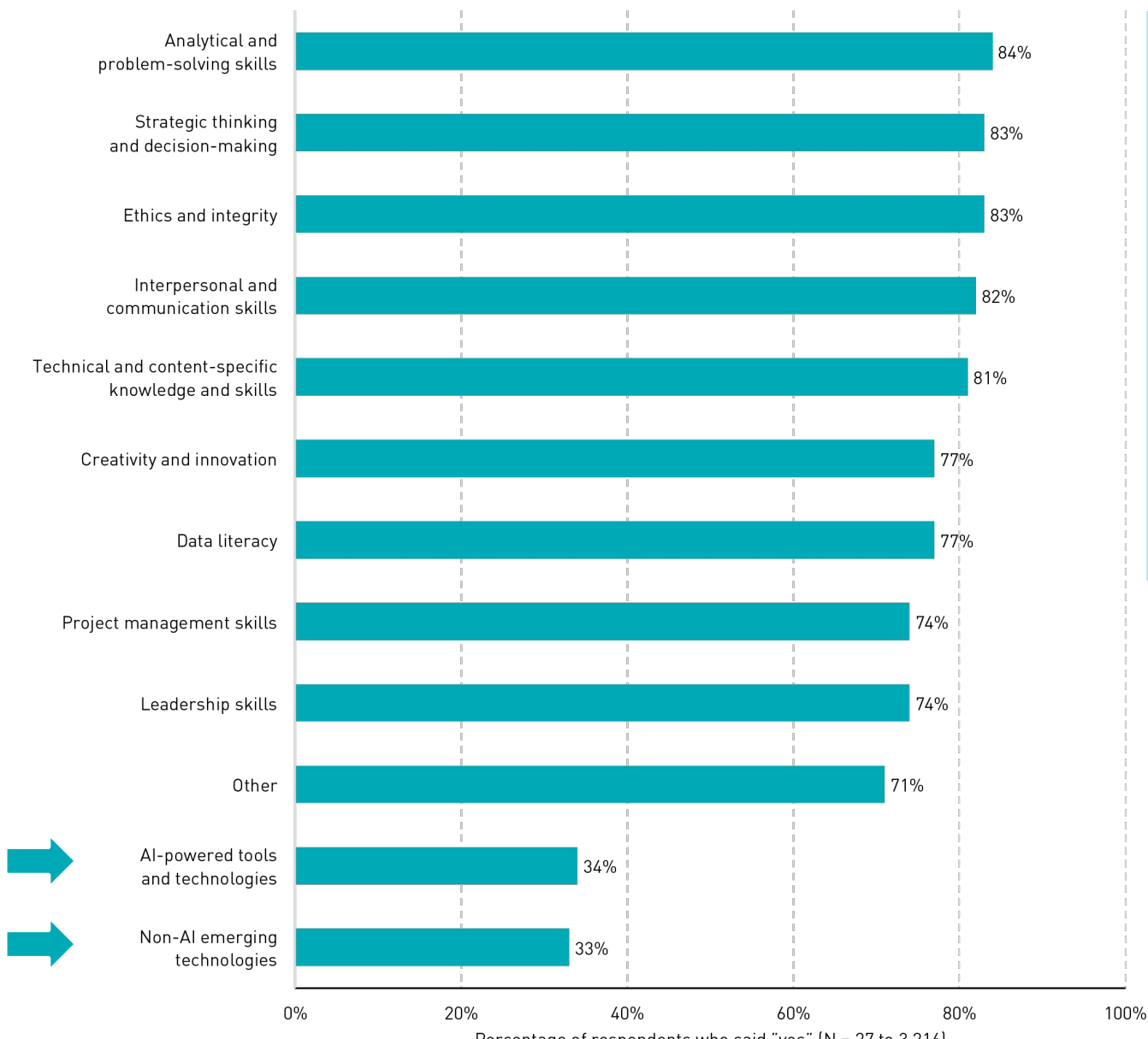


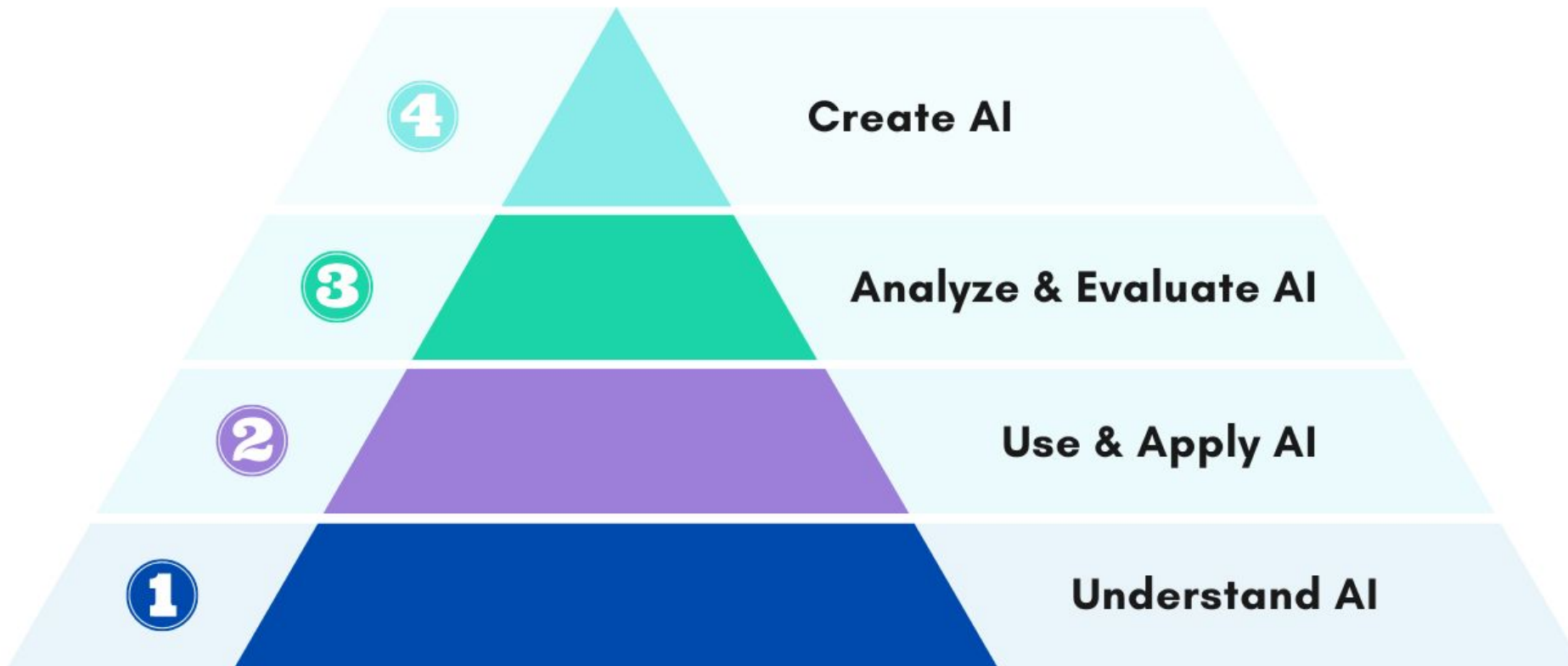
Figure 16 in EDUCAUSE 2025 Survey: Percentage of Students Who Feel Well Prepared in Career Competency Areas

Students at institutions with a cutting-edge approach to technology feel more career-ready.

Muscanell, N., & Gay, K. (2025). 2025 students and technology report: Shaping the future of higher education through technology, flexibility, and well-Being [Research report]. *EDUCAUSE*.
<https://www.educause.edu/content/2025/students-and-technology-report>

AI Literacy Building in Higher Ed

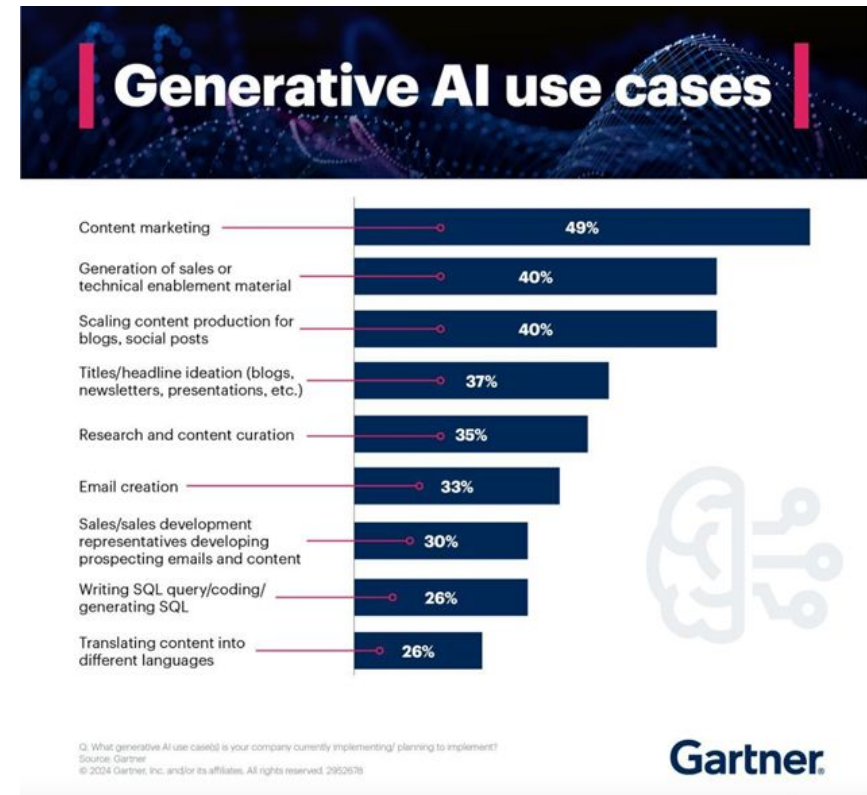
A Framework for AI Literacy (Hibbert et al., 2024)



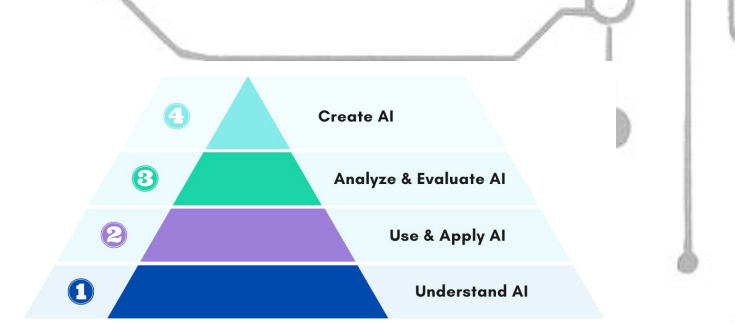
Hibbert, M., Altman, E., Shippen, T., & Wright, M. (2024, June 3). *A framework for AI literacy*. *EduCause Review*.
<https://er.educause.edu/articles/2024/6/a-framework-for-ai-literacy>

How Educators Use Gen AI for Teaching and How Businesses Use Gen AI

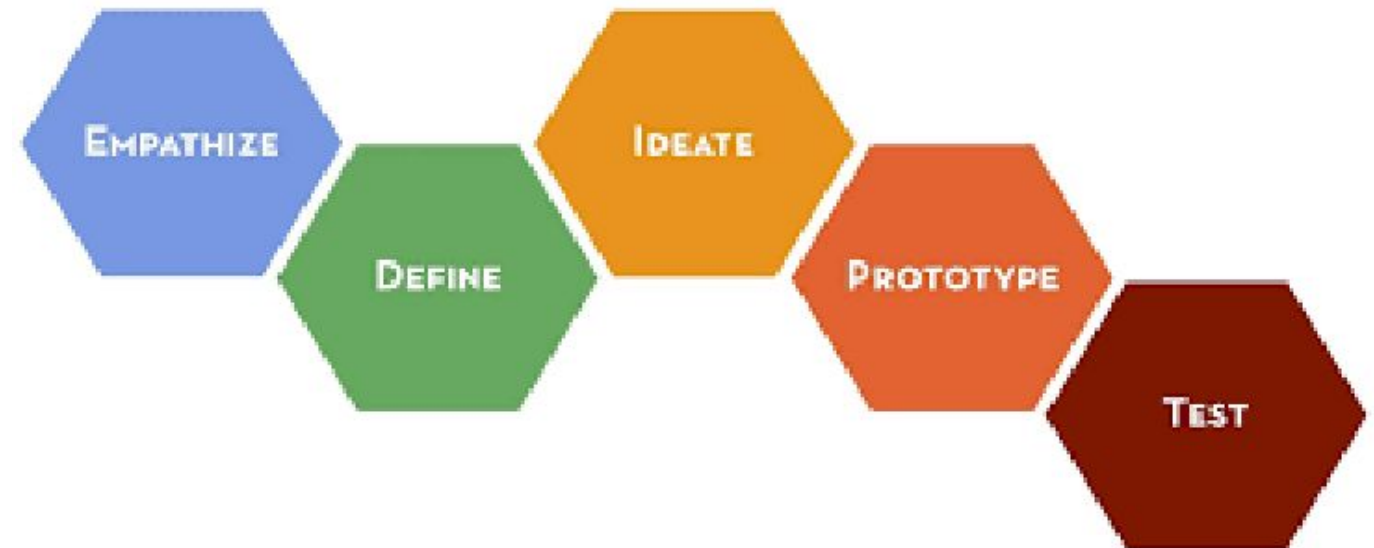
- Research
- Analysis, including a comparison of the outcomes from different gen AI tools
- Ideation and content creation
- Prototyping
- Coding
- Practicing interview and presentation skills
-



Teaching with AI Example Using AI Tools for Design Thinking



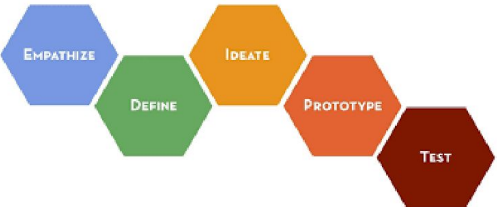
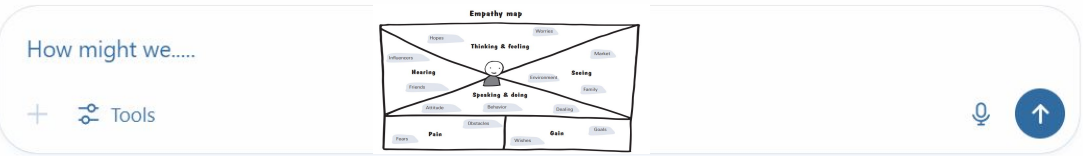
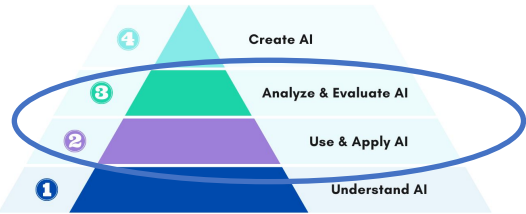
- Accelerate process simulation to complete projects during semester
- Overcome lack of access to stakeholders for interviews (personas, empathy maps)
- Apply AI/digital tools while experiencing also their limitations



• Sanchez-Persampieri, E., & Schroeder, K. (2025, May). Combining design thinking and generative AI technologies in the classroom: A project-based learning approach. *Journal of Interactive Technology and Pedagogy*, 26.
<https://cuny.manifoldapp.org/read/combining-design-thinking-and-generative-ai-technologies-in-the-classroom/section/52735f7f-8a08-4690-8926-a08ba5e8cf73>

Teaching with AI Example (Continued)

Integrating AI in Brainstorms



Idea	Desirability	Feasibility	Viability	Total Points	Source (AI or Peers)

Develop a prompt based on your empathy map and customer persona with **focus on pains and gains to generate ideas worth pursuing to solve the pain points.**

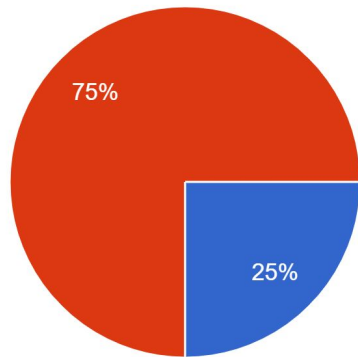
- Step 1:** Ask ChatGPT or other AI tool of your choice to provide you with 5 ideas. Enter ideas into table in Miro. Save the search to track how the idea was generated in a separate document. Tip: be specific with your prompts.
- Step 2:** Present synopsis of your identified pain points/needs. Class to brainstorm ideas for “**How Might We**” and enter 5 ideas. Save all ideas for record keeping.
- Step 3:** Identify best ideas from both options based on desirability, viability, and feasibility.
- Step 4:** Reflection about the opportunities and limitations of gen AI tools.

Student Reflections: Advantages and Limits of Using AI Tools for Design Thinking

(TMGT Class Poll, October 2024)

1. After you entered and rated all your top 10 brainstorm suggestions from gen AI and the class based on feasibility, desirability and viability, from which source was your #1 suggestion?

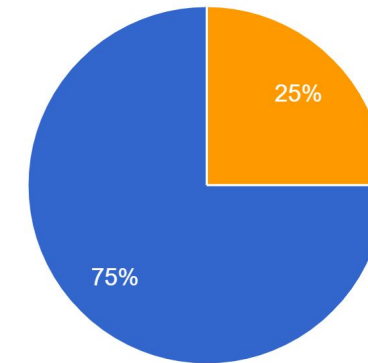
8 responses



● Top idea came from Gen AI
● Top idea came from class

2. Did you find the gen AI suggestions helpful?

8 responses



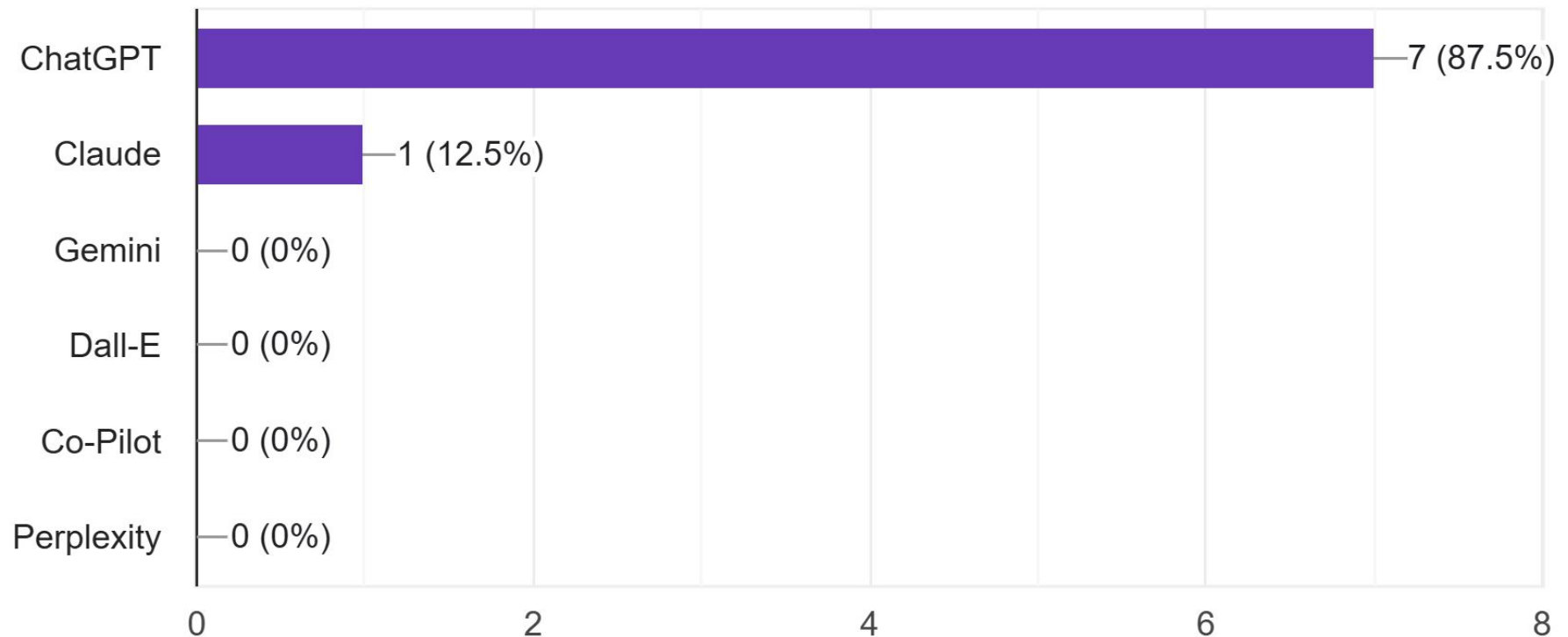
● Yes
● No
● Partially

Student Reflections : Advantages and Limits of Using AI Tools for Design Thinking

(TMGT Class Poll, October 2024)

3. Which gen AI tool did you use?

8 responses



Comments on Advantages and Limits of Using AI Tools for Design Thinking

(TMGT Class Poll, October 2024)

Limitations

- More basic ideas, no specific/applicative ideas,
- Lacked depth and practical understanding of specific customer needs, making them harder to implement
- Lacks emotion and empathy and cannot relate to the experiences that target customers have.
- AI is helpful for the feasibility and viability options, but the desirability score comes from the actual people.
- The ideas were not THAT original! It is such a reminder that we should use human ideas first and use AI to supplement
- Unsure if gen AI is capable of coming up with TRULY innovative and brand new ideas; trained on datasets of information that already exists

Advantages

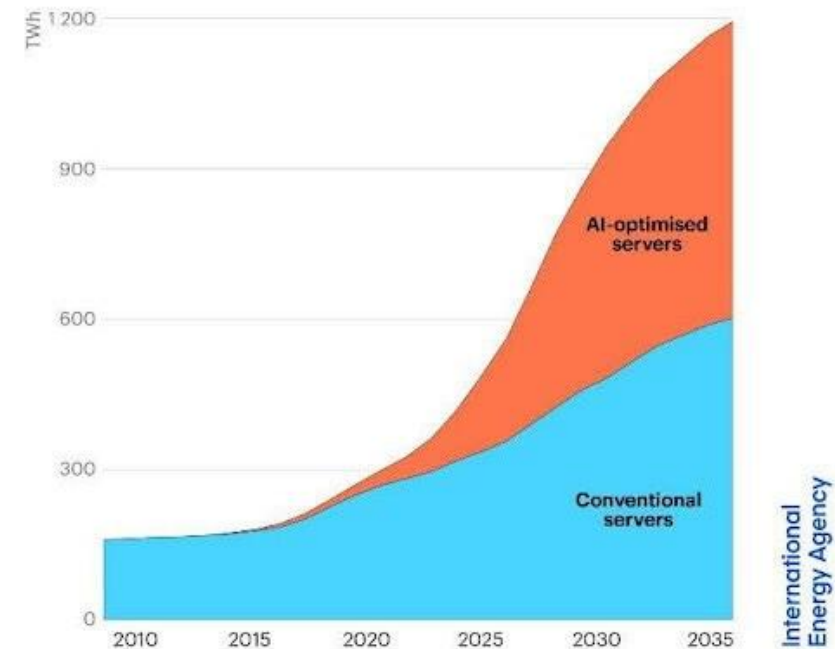
- In the near future, people will use generative AI more and more often, so it was exciting to explore its usefulness.
- It was very informative and I got some very cool ideas
- The comparison shows me that while AI can produce innovative ideas, the insights from class discussions are often more in tune with real customer needs, resulting in stronger, more relevant solutions. Also worth mentioning that 1 of the 3 top ideas come from AI.

The Role of Research/Education to Assess the Sustainability Impact of AI: #AI4Good vs #AIRisks



Data centre electricity demand is set to surge in the next decade, **driven by AI**

Data centre electricity demand, historical & projected through 2035



- Environmental Impact: increased electricity demand and water consumption.
- Social Impact: AI divide

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